

We model aggregate or industry output with a Cobb-Douglas function:

$$Y = AK^\alpha L^{1-\alpha}, \quad (1)$$

$$L = \sum_{i=1}^N \ell_i e_i, \quad (2)$$

where Y is output, K is capital, ℓ_i is hours worked by worker i , e_i is the efficiency of worker i , and L is the effective quantity of labor. If worker i saves s_i hours per week due to genAI and spends this time on additional production within their job, their effective labor becomes $\ell_i + s_i$ and the change in the effective quantity of labor is

$$\Delta L = \sum_{i=1}^N s_i e_i. \quad (3)$$

Assuming a competitive labor market and constant TFP, capital, and hours worked, Appendix D.1 shows that we can write the approximate percent change in output from genAI as:

$$\frac{\Delta Y}{Y} \approx \underbrace{(1 - \alpha)}_{\text{labor cost share}} \times \underbrace{\frac{\sum_i s_i \tilde{w}_i}{\sum_i \ell_i \tilde{w}_i}}_{\Delta \% \text{effective labor}} \quad (4)$$

where $\tilde{w}_i$ is worker i 's wage relative to mean wages. This equation states that the percent change in output due to genAI is the ratio of mean time savings to mean hours worked, weighted by worker's wages, and scaled by labor's share of production. Information on time savings and wages is available in the November 2024 RPS.

Figure 9a plots genAI use against time savings at the industry level using RPS data. We observe considerable variation in genAI use across industries, which is strongly correlated with genAI time savings. Information Services and Finance have the highest genAI use (9.4% and 9.6%, respectively) and time savings (2.7% and 2.3%, respectively), while Accommodation and Food Services has the lowest use (1.5%) and time savings (0.6%). The slope of the dashed OLS line is 0.21, indicating that a 10 percentage point increase in genAI use is associated with a 2.1% rise in time savings. The relationship is approximately linear; in particular, we do not observe a plateauing of time savings at higher rates of adoption.

Next, Figure 9b translates time savings into productivity gains using equation (4). The impact of time savings on productivity depends on two factors: (i) the share of labor in production costs, and (ii) the wages of genAI users relative to non-users within the industry. (Appendix Table D.1 lists the labor shares used for each industry.)

The implied productivity gains range from a low of 0.1% in Real Estate to a high of 1.7%

in Professional, Scientific, and Technical. The correlation between productivity gains and AI usage by industry (0.65) is smaller than for the time-savings and industry (0.93). This is primarily due to variation in labor shares across industries. For example, Information Services has high genAI use and time savings, but a low labor share (0.39).

To what extent are these predicted productivity gains evident in economic data? Figure 9c compares changes in effective labor supply in the RPS with annualized labor productivity growth (i.e., the percent change in real value added per worker) from 2022 – 2024. To adjust for trends in TFP growth prior to the Covid-19 pandemic, we detrend annualized labor productivity by subtracting the annualized growth predicted by the trend over the period 2015-2019 from the actual growth rate. We find a positive correlation of 0.39. The power of this analysis is limited by three factors. First, our predicted changes in effective labor supply are modest relative to observed changes in labor productivity, which can be affected by many factors other than genAI. Second, the change in labor productivity over this period may partly reflect recovery from the Covid-19 pandemic: in particular, both Retail and Transportation were disproportionately negatively impacted by demand shifts during the pandemic and exhibit very high productivity growth from 2022-2024 relative to trend. Third, our sample size includes only 15 industries with sufficient sample size.

Figure 9d compares changes in predicted productivity gains in the model with annualized labor productivity growth. In this comparison the correlation declines to effectively zero. Mechanically, the correlation declines because some industries with high effective labor gains and high observed productivity gains have a small labor share, which dampens the predicted productivity gain in the model. One potential explanation for the decrease in the correlation is that industry adoption of genAI may be correlated with worker adoption. If industries with large increases in effective labor supply also experienced productivity gains due to firm-level adoption, then our model, which only captures changes in individual workers’ effective labor supply, would under-predict total productivity gains in those industries.

5.4 Aggregate Productivity Gains and a Comparison of Micro and Macro Estimates

We can apply this same procedure to estimate aggregate productivity gains due to genAI. In particular, we estimate that on the aggregate level genAI currently increases the aggregate effective labor supply by 2.1%. Using an AI-exposure-adjusted labor share of 0.57, following Acemoglu (2024), we estimate a potential productivity gain of 1.2%.

To compare our estimate of the aggregate productivity gain from genAI with experimental estimates from the literature, we reformulate the model to express time savings as a linear

function of time spent using genAI, $s_i = \gamma u_i$:

$$\Delta L = \sum_{i=1}^N \gamma u_i e_i. \quad (5)$$

In this expression, u_i is weekly hours spent using genAI by worker i and γ is the productivity gain associated with one hour of genAI use.

Given our estimates of genAI use, Appendix D.1 shows that the value of γ that would generate a 2.1% increase in aggregate effective labor supply can be written as:

$$\gamma = 2.1\% \cdot \left(\frac{\sum_i \tilde{w}_i l_i}{\sum_i \tilde{w}_i u_i} \right) \quad (6)$$

As discussed in Section 5.1, RPS data provide a lower and upper bound for l_i , $\underline{l}_i$ and $\bar{l}_i$. Evaluating (6) using the midpoint of these bounds, $l_i \equiv (\underline{l}_i + \bar{l}_i)/2$ yields a wage-weighted share of total work hours spent using genAI of 5.7%, which yields $\gamma = 0.30$. This implies that each hour spent using genAI increases the worker’s productivity for that hour by 30%. This is similar in magnitude to the average productivity gain of 27% from several randomized experiments of genAI usage (Cui et al., 2024; Dell’Acqua et al., 2023; Noy and Zhang, 2023; Peng et al., 2023).

Our estimated aggregate productivity gain from genAI (1.2%) exceeds the 0.7% estimate by Acemoglu (2024) based on a similar framework. In our framework and in Acemoglu (2024), the aggregate productivity gain is the product of three numbers: the AI-exposure-adjusted labor share of GDP, the wage-weighted share of labor that is supplied by genAI, and the productivity gain from genAI. We use the same labor share as Acemoglu (2024). Our survey data imply a wage-weighted genAI adoption share of 7.0%, while Acemoglu (2024) assumes a slightly lower wage-weighted adoption share of 4.6% based on task-exposure predictions. As discussed above, we estimate a productivity gain from genAI of 30%; Acemoglu (2024) assumes a productivity gain from genAI of 27% based on micro experimental evidence. Adjusting our productivity value γ from 30% to 27% lowers our estimated aggregate productivity gain to 1.1%; the remaining 0.4 percentage point gap is due to our higher adoption estimate.

We conclude with two points about our estimate of aggregate productivity gains. First, Acemoglu (2024) assumes these productivity gains take a decade to materialize. In contrast, our estimates reflect current adoption and time savings, implying a faster pace of diffusion. Second, our estimate does not capture potential productivity gains from using genAI to reorganize or completely automate production processes, which could further increase output per worker. With previous technologies, these more systematic gains occurred with substantial time lags, implying that this omission may be modest in the short run (Bresnahan and Trajtenberg, 1995; Brynjolfsson and Hitt, 2000; David, 1990; Jovanovic and Rousseau, 2005).

6 Conclusion

Predictions about genAI’s economic impact vary widely, driven by uncertainty over the share of work it can assist and the size of its productivity potential. This paper presents evidence from the first nationally representative U.S. surveys on generative AI use at work and home. We find that a substantial share of individuals already use genAI. Thus far, the pace of overall adoption is rapid compared with past ICT technologies. At work, genAI adoption has followed a similar pace to PCs and is far ahead of genAI adoption by firms. We document wide variation in genAI use across demographic groups and occupations, and provide evidence that both task composition and firm policies appear to play an important role in workers’ adoption choices. Although the intensity of genAI use varies widely, genAI already assists a noticeable share of total work hours, suggesting the potential for meaningful productivity gains.

Our findings help lay the groundwork for understanding the labor market implications of genAI. They can also inform responses by managers and policymakers along several dimensions, such as data center and electricity needs, workforce training, and competitive dynamics.

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